

User Interfaces

Degree in Computer Engineering

2025/2026 academic year

WEB TECHNOLOGIES - HTML5 AND CSS3

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Introduction

The first block of exercises aims to familiarize yourself with the basic technologies for website development: HTML5 and CSS3. To this end, a brief introduction to the technologies will be provided, the basic syntax of both specifications will be demonstrated, a series of examples will be reviewed, and a programming exercise will be completed. This section introduces the requirements for completing the block of exercises: estimated time, due date, and evaluation requirements.

Dedication

Completing the programming exercises will require an estimated TEN HOURS of dedication per student, spread over three weeks of work and three small-classroom sessions. During the small-classroom sessions, in addition to answering general questions, the technologies involved will be explained.

Delivery

The programming exercise will be submitted during the fourth week of the semester (see the practice schedule published in Aula Global as "Practice Schedule") **before the small group session.**

Assessment

Evaluation criteria will focus on, among others: **functionality**, code , and **user interface design**.

This first set of exercises will be worth 10% of the final continuous assessment grade. Copying exercises from classmates or from the internet will result in **immediate failure** of the continuous assessment, both for the copying group and for the group that created the original version (if applicable).

Websites

A **website** is defined as a set of web pages and resources hosted on the same domain and accessed through a web browser, using *Hyper Text Transfer Protocol*.

(HTTP) as a transport protocol. A particular type of website is a **static website**. A static website is one whose content is stored on the server in a format that allows it to be sent to the client without any processing. The web pages of a static website are coded in Hypertext Markup Language (HTML), using the *Cascade Style Sheets* (CSS) specification to modify the presentation of the pages.

Hyper Text Markup Language (HTML)

HTML, an acronym for *Hyper Text Markup Language*, is a markup language oriented toward the structure and presentation of information. It is characterized by containing markup that defines the structure of the document, as well as its appearance in the display space. The main problem with HTML lies in its lack of semantics; that is, an HTML document is not "aware" of its content; it simply structures it and defines how it should be presented to the user. The HTML language is defined by the W3 consortium (www.w3c.org).

HTML5 is the fifth revision of the World Wide Web's "core" programming language, HTML. This new version aims to replace the current (X)HTML, correcting problems encountered by web developers, updating the code to meet the new needs of today's web, and adding semantics to the content with new tags.

For more information on HTML, we recommend consulting the w3schools HTML tutorial [1], which uses interactive examples to illustrate the language's essential features. The HTML language specification [2] serves as a reference and resource for advanced topics. The HTML5 specification is available at [3].

Cascade Style Sheets (CSS)

The *Cascading Style Sheets* (CSS) specification allows HTML presentation elements to be grouped into categories that facilitate their reuse and simplify HTML code. This allows presentation characteristics (colors, sizes, etc.) to be defined in CSS sheets (usually described in a separate file), and the HTML is used to define the structure (sections, subsections, etc.) of the document. Currently, there are three versions of CSS: CSS1, CSS2, and CSS3. CSS version 3 does not contradict the CSS2 specification; it merely adds functionality and refines definitions.

The fundamental objectives of the CSS specification are basically the following:

- Solve the problem that HTML has in clearly separating content and styles. HTML was originally a language designed to define the content of a document.
As attributes were added, creating document pages became more difficult. Style sheets were created starting with HTML 4.0 to clearly separate content and styles.
- Saves programmers work. Typically, a website's styles are common across all its pages. By defining these styles in a common sheet, the pages are greatly simplified.

For more information about the CSS specification, we recommend consulting the following references: First, the w3schools CSS tutorial [4], which includes a section that allows you to test various CSS elements by interactively observing the result in the browser. More elaborate examples and some demos can be seen in [5]. Finally, the official CSS1 [6] and CSS2 [7] specifications of the W3C consortium will serve as a reference for advanced topics.

Exercise

The objective of this exercise is to learn and practice using the main elements of HTML5 and CSS3. The following section presents the script, suggesting examples to use as a reference (DO NOT COPY THEM), as well as the materials provided and required for completion.

IMPORTANT: The exercise must provide the style in an external CSS style sheet, linked to the HTML file.

Statement

The website to be developed is for a collaborative site about long-distance backpacking travel. On this site, users will be able to register, log in, explore experiences shared by backpackers (accommodations, visas, transportation, en route difficulties, etc.), and purchase travel packages.

The first web page to be developed should be similar to the one shown in the image and should be composed of the following sections: header, main body (with two vertically stacked blocks—top and bottom—and a sidebar), and footer. Additionally, the following requirements must be met:

- The header should contain the website name on the left (e.g., *Backpackers Without Borders*); in the center, a horizontal navigation menu with fictitious links (*Home, Experiences, Tips, Forum, Contact*); and on the right, a logo along with a combo box that allows you to select the page language (not functional).
- The main body of the page will be made up of two containers:

The left container will occupy two-thirds of the total width of the page. It will be divided vertically into two containers: the top block and the bottom block, with the following requirements:

- ° The top block will contain a banner with a value proposition and a cover background image (e.g., backpackers traveling, mountains, etc.). Below the banner, a carousel of images with "Recommended Travel Packages" will appear, with arrows to the left and right (non-functional). Each package will have an image, description, price, and a *"Buy"* button. The *Buy* button takes you to the web page shown in **Version c**, with a purchase form.
- ° The bottom block will contain a grid of "experience cards." Each experience card will include an image, a title, a *"See More"* button, and a *"Like"* icon. The button and icon will link to the "Registration" page shown in **Version a**.

o The right container (sidebar) will take up one third of the total width occupied by the page and will contain:

• An access form composed of two fields for username and password, a “Remember credentials” *checkbox* and a “Login” button that leads to the web page shown in **Version b** (simulating that the credentials are correct).

• A “Registration” button, which will allow access to the web page shown in **Version a**, where new users can register.

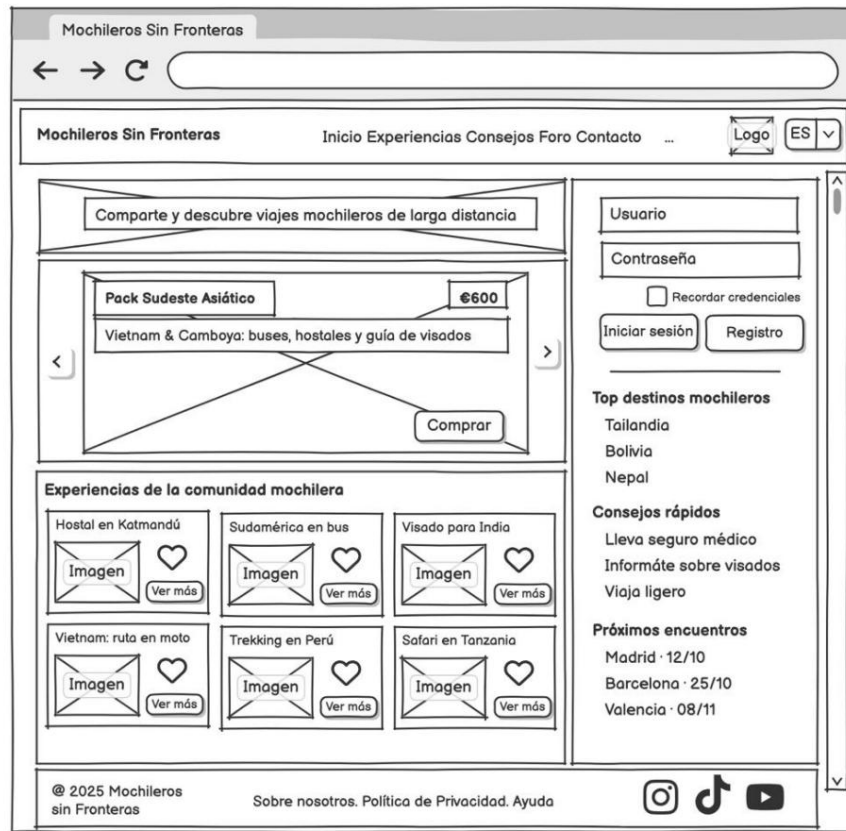
• A “*Top Backpacker Destinations*” section with a list of countries/cities (with fictitious links).

• A section with a list of “*Quick Tips*” (e.g., “Get medical insurance,” “Find out about visas in advance”; with fictitious links).

• A section with a list of “*Upcoming Meetings*”; with links fictitious.

- The footer must contain the page's copyright, three or more fictitious links (e.g., About Us, Privacy Policy, Help), social media icons (Instagram, TikTok, YouTube, or other networks of the student's choice), and a mini-newsletter subscription form (email field and a button).

ATTENTION: The image below serves as a guide to understanding the layout of the different elements. Customizing the colors, text, images, and icons of the implemented page is MANDATORY. The page content (logo, titles, images, text, links, etc.) must be authentic. Any fictitious content, such as “Lorem ipsum” text, will not be accepted.



Home (example) – desktop version

Based on the developed website, three new versions will be developed, providing the following modifications:

Version a

By clicking on the “Register” button on the previous page (in the right sidebar), you will access the **Version a** web page (see example image), where:

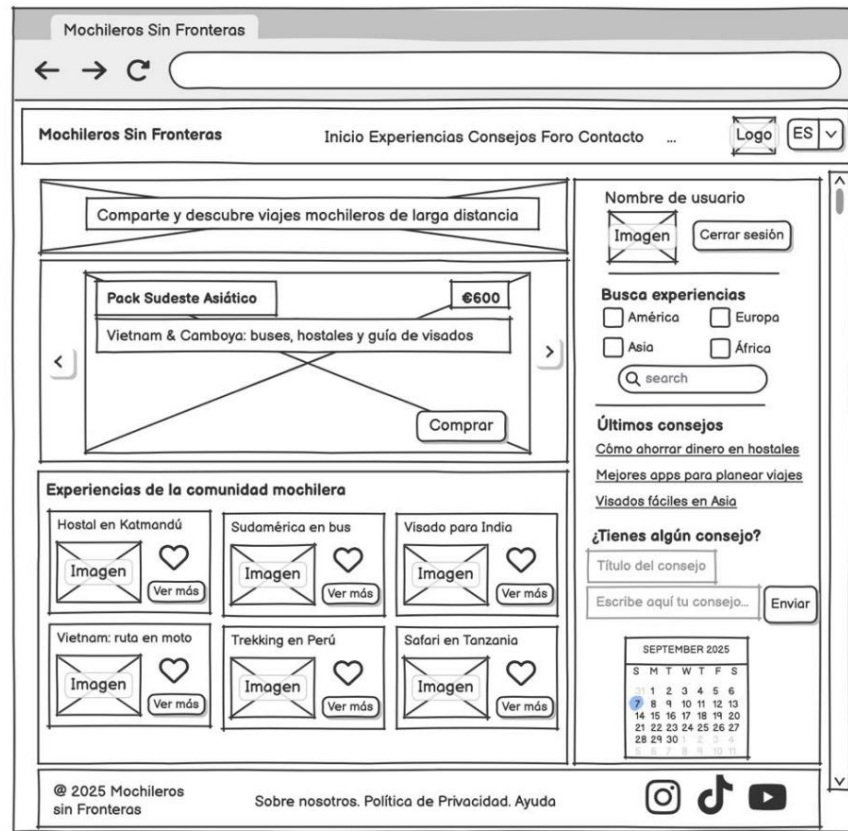
- There will be a new user registration form with the following editable fields: first name, last name, email, email confirmation, date of birth, login, password, and the text "upload profile image" accompanied by an option to upload a file. Additionally, there will be a checkbox to accept the privacy policy, accompanied by a (fictitious) link to a page with the privacy policy.
Finally, a button will appear, “Save data and access,” to submit the form.
- The “Save data and access” button must be a link. When you click on it, the page
The web that will appear will be **Version b**.

Version a (example) – desktop version

Version b

When you click on the “Save data and access” button in **Version a**, the web page that will appear will be **Version b**, which will be similar to the home page *but* with new elements in the bar. lateral. Specifically:

- Instead of the login form, the user’s name and profile picture entered in the “Registration” form (simulated) will appear, along with a (non-functional) button to log out and return to the main page.
- The “Top Backpacking Destinations” section will be replaced by a “Search Experiences” block, designed to filter “Gallery Travel Packages” and experience cards by category or keyword. It will contain *checkboxes* (at least four, e.g., with continent names) and a search bar with a magnifying glass icon for entering other keywords.
- The “Quick Tips” section will be replaced with a block called “Latest Tips,” with a list of links to the latest forum tips (simulated) and a form titled “Do you have any tips?” to enter a tip with a title and description.
- The links in the “Upcoming meetings” section will be replaced by a block with a mini-calendar (non-functional).



Version b (example) – desktop version

Version c

When you click on the "Buy" button on the carousel images, the web page that will appear will be **Version c**. This page will have two containers:

- The container on the left will display an image with the selected "Travel Package," along with a description and price.
- The container on the right will contain a purchase form. The form must include name, email address, a combo box to select the card type, card number, cardholder name, expiration date, and CVV security code. Finally, a "Buy" button will appear to submit the form, and a "Delete" button will appear to clear the form data.

Version c (example) – desktop version

In addition, the following requirements will be taken into account for all developed pages:

- The website must be *responsive*, so that it adapts to the browser size as it changes. To achieve this, at least two *breakpoints must be implemented*: one for changing to *tablet* (768px) and another for changing to *mobile* (600px). In the *tablet* and *mobile* versions, the layout of the different elements and their size will change depending on the available space. Students will be responsible for deciding how to arrange the different elements in each of the versions.
- All website content (logo, title, images, text, links, etc.) must be realistic and thematic (e.g. “Lorem Ipsum” text will not be accepted).
- Semantic labels should be used appropriately to represent the different elements. In addition, only HTML and CSS code can be used.
- The delivered code must successfully pass the W3C Markup Validation Service (<https://validator.w3.org/>).

To solve the exercise, we recommend consulting the following HTML and CSS explanations from the w3schools website (www.w3schools.com/html/default.asp): _____

HTML Layouts – Examples of how to use the DIV element to structure documents
HTML.

HTML Images – Examples for handling images in HTML documents.

HTML Links – Examples of how to include links to other pages.

HTML Styles – Examples of how to style HTML elements.

HTML Forms – Examples of how to include forms in HTML.

CSS Floating and CSS Position – Explanation of element positioning.

CSS Display – Explanation on how to show/hide an element.

CSS Margin – Margin Explained.

CSS Padding – Padding Explained.

CSS Navigation Bar – Explanation of the navigation menu.

CSS width/height property – Explanation of the width and height properties.

CSS Image Gallery – An explanation of how to create an image gallery.

RWD Viewport – Explanation of the viewport property.

RWD Media Queries – Explanation of how to use media queries.

RWD Images – Explanation of using properties to make images responsive.

Required material

To complete the exercises, you may not use any HTML page editor, framework, or specific tool, such as Dreamweaver. However, we recommend using a free editor such as Notepad++, HTML-Kit, Visual Studio Code, Sublime Text, Brackets, or Atom. The instructors will not provide support for these tools.
tools.

Regulations

The execution and submission of the programming exercise will be governed by the following regulations. Failure to comply with any of these rules will result in the inability to obtain a **grade higher than three** in the programming exercises block.

Implementation standards

The resolution of the programming exercise must comply with the following rules:

- The exercise must be performed in groups of two. Group members must be from the same practice group and cannot change during practice blocks.
- The exercise will be solved using valid HTML 5 and CSS 3 code.
- The exercise should be tested in Firefox and Chrome, ensuring that it looks and works correctly.

Delivery rules

The programming exercise must be submitted on the day of the session indicated in the introduction to this statement. The submission must comply with the following rules:

- Submission will be made through Aula Global. A single compressed .zip or .rar file named ep01_grXX must be submitted, where **XX** corresponds to the practice group identifier. For example, practice group 5 of group 81 must generate a file named:

ep01_gr05.zip or ep01_gr05.rar

The folder structure of the files submitted must be as follows:

- **e.g.** Exercise root directory. HTML files.
- **ej/style.** Style sheets.
- **ej/images.** Images and graphic resources.

References

- [1] "*HTML Tutorial*", W3 Schools HTML Tutorial, available at <http://www.w3schools.com/html/>
- [2] "*HTML 4.01 Specification*", W3C recommendation, available at <http://www.w3.org/TR/html4>
- [3] "HTML5", W3C recommendation, available at <http://www.w3.org/TR/html5/>
- [4] "*CSS Tutorial*", W3 Schools CSS Tutorial, available at <http://www.w3schools.com/css/>
- [5] "CSSPlay", CSS demos and examples, available at <http://www.cssplay.co.uk/index.html>
- [6] "*Cascading Style Sheets, level 1*", W3C specification, available at <http://www.w3.org/TR/REC-CSS1>
- [7] "*Cascading Style Sheets, level 2*", W3C specification, available at <http://www.w3.org/TR/REC-CSS2>